Heavy-Ion Collision Animation

Every formula used in the code

heavyIonCollision project

September 26, 2026

This document lists every formula the software evaluates, in the order of the pipeline: geometry → initial state → fireball → hadrons → scene. Each block names the file that implements it and the literature source (full references in `docs/physics/01--04`). Two markers flag what is *not* textbook physics: [E] = empirical or calibration choice made in this project (tuned so that the output matches data, see `docs/validation.md`); [V] = purely visual mapping (does not affect any physical number).

Units. Natural units $\hbar = c = k_B = 1$ unless stated: lengths in fm, times in fm/c, energies and momenta in GeV, temperatures in MeV where noted.

$$\hbar c = 0.1973269804 \text{ GeV fm}, \qquad 1 \text{ fm}/c = 3.33564 \times 10^{-24} \text{ s},$$
$$m_p = 0.93827208943 \text{ GeV}, \quad m_n = 0.93956542194 \text{ GeV}, \quad m_N = \tfrac{1}{2}(m_p + m_n).$$

[`hic_core/constants.py`; PDG 2024]

Contents

1 Beam kinematics **2**

2 Nuclear density and nucleon sampling **2**

2.1 Woods–Saxon (2pF / 3pF) with deformation . . . 2

2.2 Sampling . . . 2

2.3 Light nuclei . . . 3

3 Monte Carlo Glauber model **3**

4 Initial entropy (TRENTo) **4**

4.1 Multiplicity for systems without their own table[E] . . . 4

5 Equation of state (lattice QCD) **4**

6 Fireball evolution (Level A) **4**

6.1 Longitudinal profile . . . 4

6.2 Transverse expansion . . . 5

6.3 Flow field . . . 5

6.4 Temperature . . . 5

7 Hadrons: formation, freeze-out and emission **6**

7.1 Number and species 6
7.2 Fluid tracers 6
7.3 Cooper–Frye / blast-wave emission 6
7.4 Observables 7
7.5 Positions at lab time t 7

8 Data lookup and calibration **7**

9 Nucleons, spectators and the time axis **7**

10 Importers (Level B) **8**

11 Special functions **8**

12 Visual mappings in Blender[V] **9**

13 Validation metric **10**

14 Interactive explorer additions (The Little Bang) **10**

Multi-nucleon transfer explorer **10**

15 Nuclear input data **10**

16 Nucleus–nucleus potential **11**

17 Classical trajectory with surface friction **11**

18 Nucleon exchange **12**

19 Cross sections, kinematics and de-excitation **12**

1 Beam kinematics

[hic_core/kinematics.py]

Lorentz factor, velocity and beam rapidity of a nucleon in the collider (centre-of-mass) frame:

$$\gamma = \frac{\sqrt{s_{NN}}}{2m_N}, \qquad \beta = \sqrt{1-\gamma^{-2}}, \qquad y_{\mathrm{beam}} = \operatorname{arccosh}\gamma. \tag{1}$$

Longitudinal thickness of a contracted nucleus and the time the two discs need to cross:

$$L_z = \frac{2R}{\gamma}, \qquad t_{\mathrm{cross}} = \frac{L_z}{\beta}. \tag{2}$$

Inelastic nucleon–nucleon cross section: tabulated value when $\sqrt{s_{NN}}$ is within 1% of a pre-set (PHENIX PRC 93 024901; Loizides et al.), otherwise the fit of d'Enterria & Loizides (arXiv:2011.14909), with s in GeV^2:

$$\sigma_{NN}^{\mathrm{inel}}(s) = a + b\,\ln^n s, \qquad a = 28.84\ \mathrm{mb},\ b = 0.0458\ \mathrm{mb},\ n = 2.374. \tag{3}$$

Black-disk interaction distance (1 mb = 0.1 fm^2):

$$D = \sqrt{\sigma_{NN}^{\mathrm{inel}}/\pi}. \tag{4}$$

Milne coordinates (proper time and space-time rapidity) and their inverse:

$$\tau = \sqrt{t^2 - z^2}, \quad \eta_s = \operatorname{artanh}\frac{z}{t}; \qquad t = \tau\cosh\eta_s, \quad z = \tau\sinh\eta_s. \tag{5}$$

2 Nuclear density and nucleon sampling

[hic_core/nuclei.py; arXiv:1408.2549; arXiv:1710.07098]

2.1 Woods–Saxon (2pF / 3pF) with deformation

$$\frac{\rho(r,\theta)}{\rho_0} = \frac{1 + w\,r^2/R^2}{1 + \exp\big[(r - R(\theta))/a\big]}, \qquad R(\theta) = R\big[1 + \beta_2 Y_{20}(\theta) + \beta_4 Y_{40}(\theta)\big], \tag{6}$$

$$Y_{20} = \sqrt{\tfrac{5}{16\pi}}\,(3\cos^2\theta - 1), \qquad Y_{40} = \tfrac{3}{16\sqrt{\pi}}\,(35\cos^4\theta - 30\cos^2\theta + 3). \tag{7}$$

Parameters $(R, a, w, \beta_2, \beta_4)$ per nucleus are in `hic_core/data/nuclei.json`. For ^{208}Pb, protons and neutrons are drawn from separate profiles (neutron skin): $(R_p, a_p) = (6.68, 0.447)$, $(R_n, a_n) = (6.69, 0.560)$ fm.

2.2 Sampling

Radius and polar angle by rejection from $r^2\rho(r, \cos\theta)$ on $r \in [0, r_{\max}]$, $\cos\theta \in [-1, 1]$, azimuth uniform:

$$r_{\max} = R\,(1 + 0.7|\beta_2| + 0.4|\beta_4|) + 10a, \qquad \text{envelope} = 1.05\,\max\big[r^2\rho\big] \text{ (on a } 400 \times 41 \text{ grid)}. \tag{8}$$

Hard-core condition: a candidate is rejected if $|\vec{x} - \vec{x}_j| < d_{\min}$ for any placed nucleon j (default $d_{\min} = 0.4$ fm). Afterwards the nucleus is recentred, $\vec{x}_i \to \vec{x}_i - \langle\vec{x}\rangle$.

Orientation of deformed nuclei (symmetry axis initially along z): random = uniform rotation from a unit quaternion $q = (w, x, y, z)$, $q \sim \mathcal{N}(0,1)^4/\|\cdot\|$,

$$\mathsf{R}(q) = \begin{pmatrix} 1-2(y^2+z^2) & 2(xy - zw) & 2(xz + yw) \\ 2(xy + zw) & 1-2(x^2+z^2) & 2(yz - xw) \\ 2(xz - yw) & 2(yz + xw) & 1-2(x^2+y^2) \end{pmatrix}; \tag{9}$$

tip-tip $=\nparallel$; body-body ($z \to y$) and body-body along b ($z \to x$):

$$\mathsf{R}_{\mathrm{bb}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad \mathsf{R}_{\mathrm{bb,x}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}. \tag{10}$$

2.3 Light nuclei

Deuteron (Hulthén wave function, $a = 0.228$ fm^{-1}, $b = 1.18$ fm^{-1}): proton–neutron separation r from

$$P(r)\,dr \propto \left(e^{-ar} - e^{-br}\right)^2 dr, \quad r \in [0, 20] \text{ fm}; \qquad \vec{x}_{p,n} = \pm\tfrac{r}{2}\,\hat{n},\ \hat{n} \text{ isotropic}. \tag{11}$$

^{3}He (approximation[E]; published MCs use GFMC configurations): Gaussian with the charge radius $r_{\mathrm{rms}} = 1.966$ fm,

$$x, y, z \sim \mathcal{N}\big(0,\ \sigma^2\big), \quad \sigma = r_{\mathrm{rms}}/\sqrt{3}. \tag{12}$$

3 Monte Carlo Glauber model

[`hic_core/glauber.py`; Miller et al. Ann. Rev. 57 (2007) 205; LKD18]

Impact parameter ($\mathrm{d}N/\mathrm{d}b \propto b$) and the shift of the nuclei:

$$b = b_{\max}\sqrt{U},\ U \sim \mathcal{U}(0,1), \quad b_{\max} = 1.2(R_A + R_B) + 6 \text{ fm}; \qquad x_A \to x_A - \tfrac{b}{2}, \quad x_B \to x_B + \tfrac{b}{2}. \tag{13}$$

Collision criterion for a pair with transverse distance d_{ij}: black disk, or the Gaussian profile of LKD18 App. B (Q = regularised upper incomplete gamma function, Sec. 11):

$$P_{\mathrm{coll}} = \Theta(D^2 - d_{ij}^2) \qquad \text{or} \qquad P_{\mathrm{coll}} = Q\left(\frac{1}{\omega},\ \frac{d_{ij}^2}{D^2\omega}\right),\ \omega = 0.4. \tag{14}$$

$$N_{\mathrm{coll}} = \sum_{i\in A, j\in B} \mathrm{hit}_{ij}, \qquad N_{\mathrm{spec}} = A + B - N_{\mathrm{part}},$$
$$N_{\mathrm{part}} = \#\{i : \textstyle\sum_j \mathrm{hit}_{ij} > 0\} + \#\{j : \sum_i \mathrm{hit}_{ij} > 0\}. \tag{15}$$

Participant eccentricities and event-plane angles (r^n weighting, TGlauberMC/LKD18 convention; positions relative to the participant centroid):

$$\varepsilon_n = \frac{\left|\langle r^n e^{in\phi}\rangle\right|}{\langle r^n\rangle}, \qquad \Psi_n = \frac{\operatorname{atan2}\big(\langle r^n \sin n\phi\rangle, \langle r^n \cos n\phi\rangle\big) + \pi}{n}. \tag{16}$$

Overlap area:

$$S = 4\pi\sqrt{\sigma_x^2\sigma_y^2 - \sigma_{xy}^2}. \tag{17}$$

Hadronic cross section from the Monte Carlo and centrality from impact parameter:

$$\sigma_{AB} = \pi b_{\max}^2 \frac{N_{\mathrm{inel}}}{N_{\mathrm{events}}}, \qquad [b_{\mathrm{lo}}, b_{\mathrm{hi}}](c_1, c_2) = \big[\,Q_b(c_1),\, Q_b(c_2)\,\big], \tag{18}$$

where $Q_b(c)$ is the c-quantile of b among inelastic events (LKD18 tables are used directly where available). For a given b without a table[E]: $c \simeq 100\,\pi b^2/\sigma_{AB}$ (%), $\sigma_{AB} = 700$ fm^2 if unknown.

4 Initial entropy (TRENTo)

[hic_core/fireball.py; PRC 92 (2015) 011901; Nature Phys. 15 (2019) 1113]

Participant thickness with Gaussian nucleons of width w and Γ-distributed weights (k = shape):

$$T_A(x,y) = \sum_{i\in\mathrm{part}(A)} \gamma_i \,\frac{1}{2\pi w^2}\,\exp\!\left[-\frac{(x-x_i)^2+(y-y_i)^2}{2w^2}\right], \qquad \gamma_i \sim \Gamma(k, 1/k). \tag{19}$$

Reduced thickness:

$$T_R(p; T_A, T_B) = \left(\frac{T_A^p + T_B^p}{2}\right)^{1/p}, \qquad T_R(0) = \sqrt{T_A T_B}. \tag{20}$$

Defaults (Bayesian calibration): $p = 0$, $w = 0.96$ fm, $k = 1/\sigma^2_{\mathrm{fluct}} = 1.23$. Grid: 96×96 cells over $[-16, 16]$ fm. Normalisation to the measured multiplicity via the entropy per charged hadron (Hanus, Mazeliauskas, Reygers PRC 100 (2019) 064903):

$$\frac{\mathrm{d}S}{\mathrm{d}\eta} = \frac{S}{N_{\mathrm{ch}}}\,\frac{\mathrm{d}N_{\mathrm{ch}}}{\mathrm{d}\eta}\Big|_{\eta=0}, \quad \frac{S}{N_{\mathrm{ch}}} = 6.7; \qquad s_0(x,y) = \frac{\mathrm{d}S/\mathrm{d}\eta}{\tau_0\,\Delta x^2 \sum T_R}\,T_R(x,y). \tag{21}$$

4.1 Multiplicity for systems without their own table[E]

[hic_core/event.py, scripts/calibrate_trento_norm.py]

$$I = \Delta x^2 \sum_{x,y} T_R, \qquad \mathcal{N}_{\mathrm{ref}}(c) = \frac{\langle \mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta\rangle_c}{\langle I\rangle_c}, \qquad \frac{\mathrm{d}N_{\mathrm{ch}}}{\mathrm{d}\eta} = \overline{\mathcal{N}}_{\mathrm{ref}}\, I \left(\frac{\sqrt{s_{NN}}}{\sqrt{s_{\mathrm{ref}}}}\right)^{0.31}, \tag{22}$$

with $\overline{\mathcal{N}}_{\mathrm{ref}}$ the mean over reference bins with $c \le 60\%$ (Pb–Pb 2.76/5.02 TeV, Au–Au 200 GeV) and the exponent from $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta \propto s^{0.155}$ (ALICE PRL 116 (2016) 222302).

5 Equation of state (lattice QCD)

[hic_core/eos.py; HotQCD PRD 90 (2014) 094503, PLB 795 (2019) 15]

From the tabulated dimensionless ratios (T in GeV inside the powers):

$$\varepsilon(T) = \left[\frac{\varepsilon}{T^4}\right](T)\,\frac{T^4}{(\hbar c)^3}, \quad p(T) = \left[\frac{p}{T^4}\right](T)\,\frac{T^4}{(\hbar c)^3}, \quad s(T) = \left[\frac{s}{T^3}\right](T)\,\frac{T^3}{(\hbar c)^3}, \quad c_s^2(T). \tag{23}$$

ε/T^4 is interpolated linearly in $\ln T$; below 130 MeV it is held at 2.33 (display extrapolation[E]), above 400 MeV at 13.34. The inverse $T(\varepsilon)$ is a log–log interpolation of the same table. $T_c = 156.5$ MeV. Phase bands of the table: hadron gas $T < 145$, crossover $145 \le T \le 163$, QGP $T > 163$ MeV.

6 Fireball evolution (Level A)

[hic_core/fireball.py]

6.1 Longitudinal profile

Boost-invariant in shape, Gaussian in space-time rapidity, cut at the beam rapidity:

$$s(\tau, x, y, \eta_s) = f(\eta_s)\, s_{2D}(\tau, x, y), \qquad f(\eta_s) = e^{-\eta_s^2/2\sigma_\eta^2}\,\Theta(y_{\mathrm{beam}} - |\eta_s|), \tag{24}$$

$$\sigma_\eta = \sqrt{\max(\sigma_y^2 - 0.8^2,\, 1)}^{\,[E]}, \qquad \sigma_y = \begin{cases} 3.86 & \text{LHC (ALICE PLB 726 610)} \\ \mathrm{FWHM}/2.3548 & \text{RHIC (PHOBOS)} \end{cases} \tag{25}$$

where 0.8 removes the thermal smearing between η_s and rapidity.

6.2 Transverse expansion

Entropy conservation of an ideal fluid in Milne coordinates with a prescribed velocity field $\vec{v}(\tau, x, y)$, plus a small diffusion[E] ($D = 0.15$ fm):

$$\partial_\tau q + \nabla_\perp \cdot (q\,\vec{v}) = D\,\nabla_\perp^2 q, \qquad q = \tau\, s\, \gamma_\perp, \quad \gamma_\perp = (1 - v^2)^{-1/2}. \tag{26}$$

Discretisation: donor-cell upwind flux at cell faces, $u_{i+1/2} = \frac{1}{2}(v_i + v_{i+1})$,

$$F_{i+1/2} = \begin{cases} q_i\, u_{i+1/2} & u_{i+1/2} > 0 \\ q_{i+1}\, u_{i+1/2} & \text{otherwise} \end{cases}, \quad q_i^{n+1} = q_i^n - \frac{\Delta\tau}{\Delta x}\big(F_{i+1/2} - F_{i-1/2}\big) + \Delta\tau\, D\,(\nabla^2 q)_i, \tag{27}$$

$(\nabla^2 q)_{ij} = (q_{i+1,j} + q_{i-1,j} + q_{i,j+1} + q_{i,j-1} - 4q_{ij})/\Delta x^2$; $\Delta\tau = \min(0.02,\ 0.9\Delta x/\beta_s)$ fm/c; open boundary ($q = 0$ on the edge); snapshots every 0.1 fm/c. Before τ_0 (display only): $s_{2D}(\tau < \tau_0) = s_0\,\tau_0/\tau$.

6.3 Flow field

Surface velocity from the blast-wave fit ($\langle\beta_T\rangle = 2\beta_s/(2+n)$, Schnedermann–Sollfrank–Heinz), with a calibrated gain g[E] (Sec. 8) and a build-up time:

$$\beta_s = \min\Big(\langle\beta_T\rangle\frac{2+n}{2}\,g,\ 0.97\Big), \qquad v_s(\tau) = \beta_s \tanh\frac{\max(\tau - \tau_0, 0)}{\tau_{\rm acc}},$$
$$\tau_{\rm acc} = \tau_{\rm acc}^{(0)}\,\max\Big(\frac{R_0}{6.5\ \text{fm}}, 0.2\Big)^{1.5}\text{[E]}, \qquad \tau_{\rm acc}^{(0)} = 2.5\ \text{fm}/c. \tag{28}$$

Flow radius, chosen so that the entropy-weighted mean of $(r/R)^n$ equals the uniform-disk blast-wave value $2/(2+n)$:

$$R(\tau) = \left[\frac{2+n}{2}\,\frac{\sum s\, r^n}{\sum s}\right]^{1/n}. \tag{29}$$

Entropy-field eccentricities (around the entropy centroid), used for the anisotropic radius:

$$\varepsilon_n^{(s)} = \frac{\big|\sum s\, r^n e^{in\phi}\big|}{\sum s\, r^n}, \qquad \Psi_n^{(s)} = \frac{\operatorname{atan2}(\sum s r^n \sin n\phi, \sum s r^n \cos n\phi) + \pi}{n}, \tag{30}$$

$$R(\phi) = R\Big[1 - \tfrac{1}{2}\varepsilon_2 \cos 2(\phi - \Psi_2) - \tfrac{1}{2}\varepsilon_3 \cos 3(\phi - \Psi_3)\Big]. \tag{31}$$

Velocity (radial direction $\hat{r}$), with the azimuthal modulation of the transverse flow rapidity[E] that produces v_n (amplitudes a_n calibrated, Sec. 8):

$$|\vec{v}|_0 = v_s(\tau)\,\min\Big(\frac{r}{R(\phi)}, 1\Big)^n,$$
$$|\vec{v}| = \tanh\Big[\operatorname{artanh}\big(\min(|\vec{v}|_0, 0.99)\big)\ \operatorname{clip}\Big(1 + 2\sum_{m=2}^{4} a_m \cos m(\phi - \Psi_m),\ 0.05,\ 3\Big)\Big]. \tag{32}$$

6.4 Temperature

From the lattice entropy density, $s(T)$ tabulated on $T \in [20, 1200]$ MeV (with s/T^3 clamped to its 130–400 MeV values) and inverted by interpolation in $\ln s$; $T = 0$ below the grid:

$$T(\tau, x, y, \eta_s) = s^{-1}\big(f(\eta_s)\, s_{2D}(\tau, x, y)\big). \tag{33}$$

Lab-frame field at time t: $\tau = \sqrt{t^2 - z^2}$, $\eta_s = \operatorname{artanh}(z/t)$, zero outside $|z| < t$. QGP entropy fraction and the end of the QGP phase:

$$F_{\rm QGP}(\tau) = \frac{\sum s\,\Theta(T - T_c)}{\sum s}, \qquad \tau_{\rm QGP} = \min\{\tau : F_{\rm QGP}(\tau) < 0.1\}. \tag{34}$$

7 Hadrons: formation, freeze-out and emission

[hic_core/event.py, hic_core/freezeout.py]

7.1 Number and species

Total number of hadrons (all species, all rapidities) from entropy conservation:

$$N_{\rm had} = \frac{\mathrm{d}S/\mathrm{d}\eta \int f(\eta_s)\,\mathrm{d}\eta_s}{(S/N_{\rm ch})\, f_{\rm ch}\, J}, \qquad J = \frac{N_{\rm ch}(|\eta| < 0.5)}{N_{\rm ch}(|\eta_s| < 0.5)}, \tag{35}$$

$f_{\rm ch}$ = charged fraction of the species mix, J = pseudorapidity Jacobian from a 2×10^4-particle blast-wave pre-sample ($v = \beta_s\, r^n$, $r = \sqrt{U}$, $\eta_s \sim \mathcal{U}(-1.5, 1.5)$). Species fractions from measured $\mathrm{d}N/\mathrm{d}y$ (ALICE, STAR; `data/yields.json`), neutral partners by isospin and antibaryons from $r = \bar{p}/p$ (interpolated in $\ln\sqrt{s_{NN}}$):

$$\pi^0 = \tfrac{1}{2}(\pi^+ + \pi^-), \quad K^0 = K^+ + K^-, \quad n = p, \quad \bar{n} = \bar{p}, \quad \Lambda = \Big(\frac{\Lambda}{\pi}\Big)\pi^+, \tag{36}$$

$$\bar{\Lambda} = \Lambda\, r^{2/3}, \quad \bar{\Xi}^+ = \Xi^-\, r^{1/3}, \quad \bar{\Omega}^+ = \Omega^-, \quad \bar{d} = d\, r^2. \tag{37}$$

7.2 Fluid tracers

Initial positions drawn $\propto s_0(x, y)$ (cell + uniform offset), $\eta_s \sim \mathcal{N}(0, \sigma_\eta^2)$ truncated at $y_{\rm beam}$. Midpoint (RK2) advection with the flow plus the random walk equivalent to the diffusion term:

$$\vec{x}_{k+1} = \vec{x}_k + \Delta\tau\, \vec{v}\big(\tau_k + \tfrac{\Delta\tau}{2},\ \vec{x}_k + \tfrac{\Delta\tau}{2}\vec{v}(\tau_k, \vec{x}_k)\big) + \sqrt{2D\Delta\tau}\ \vec{\xi}, \qquad \vec{\xi} \sim \mathcal{N}(0, \mathbb{1}). \tag{38}$$

Hadronisation and kinetic freeze-out proper times of each tracer:

$$\tau_c = \min\{\tau_k : T < T_c\}, \qquad \tau_{f,T} = \min\{\tau_k : T < T_{\rm kin}\}, \tag{39}$$

$$\tau_f = \max\Big(\tau_c,\ \min\big(\tau_{f,T},\ \tau_c + \tau_{\rm had},\ \tau_{\rm end}\big)\Big), \qquad \tau_{\rm had} = \max\Big(\tau_{\rm had}^{(0)} \min\big(\tfrac{R_0}{6.5}, 1\big),\ 0.5\Big)^{\rm E}, \tag{40}$$

$\tau_{\rm had}^{(0)} = 4.5$ fm/c (ALICE K*/K: 4–7 fm/c). Freeze-out point in the lab: $t_f = \tau_f \cosh\eta_s$, $z_f = \tau_f \sinh\eta_s$, transverse position $\vec{x}_f$ and flow $\vec{v}_f = \vec{v}(\tau_f, \vec{x}_f)$.

7.3 Cooper–Frye / blast-wave emission

Momentum in the local rest frame (Boltzmann, isochronous surface):

$$\frac{\mathrm{d}N}{\mathrm{d}p} \propto p^2 E\, e^{-E/T_{\rm kin}}, \qquad E = \sqrt{p^2 + m^2}, \quad \hat{p} \text{ isotropic}. \tag{41}$$

Sampling: kinetic energy $K = E - m$ from the envelope $E^2 e^{-E/T}$ = mixture $m^2 T\,\Gamma(1, T) + 2mT^2\,\Gamma(2, T) + 2T^3\,\Gamma(3, T)$, accepted with probability p/E. Two Lorentz boosts, first by the transverse flow $\vec{\beta} = (\vec{v}_f, 0)$, then longitudinally by $\vec{\beta} = (0, 0, \tanh\eta_s)$:

$$E' = \gamma\big(E + \vec{\beta}\cdot\vec{p}\big), \qquad \vec{p}\,' = \vec{p} + \Big[\frac{(\gamma - 1)\,\vec{\beta}\cdot\vec{p}}{\beta^2} + \gamma E\Big]\vec{\beta}. \tag{42}$$

This is the Monte Carlo form of the Schnedermann–Sollfrank–Heinz blast wave

$$\frac{\mathrm{d}N}{m_T\,\mathrm{d}m_T} \propto \int_0^R r\,\mathrm{d}r\ m_T\, I_0\Big(\frac{p_T \sinh\rho}{T_{\rm kin}}\Big)\, K_1\Big(\frac{m_T \cosh\rho}{T_{\rm kin}}\Big), \qquad \rho = \operatorname{artanh}\beta_T(r), \tag{43}$$

which is *not* evaluated directly (the Bessel functions of Sec. 11 are implemented and tested for it).

7.4 Observables

$$p_T = \sqrt{p_x^2 + p_y^2}, \quad \phi = \operatorname{atan2}(p_y, p_x), \quad y = \tfrac{1}{2}\ln\frac{E+p_z}{E-p_z}, \quad \eta = \operatorname{artanh}\frac{p_z}{|\vec{p}|}, \tag{44}$$

$$v_n = \left\langle \cos n(\phi - \Psi_n) \right\rangle_{\text{charged, } |\eta|<0.8,\ 0.2<p_T<5\ \text{GeV}} \quad \text{(ALICE acceptance)}. \tag{45}$$

7.5 Positions at lab time t

Per hadron, $\tau_p = t/\cosh\eta_s$. Fluid phase ($\tau_0 \le \tau_p < \tau_f$): transverse position interpolated linearly in τ from the tracer history, $z = \tau_p \sinh\eta_s$. Free streaming ($\tau_p \ge \tau_f$):

$$\vec{x}(t) = \vec{x}_f + \vec{v}\,(t - t_f), \qquad \vec{v} = \vec{p}/E. \tag{46}$$

8 Data lookup and calibration

[hic_core/event.py, hic_core/freezeout.py]

Tabulated inputs are interpolated in centrality: multiplicity and yields in log, blast-wave parameters and v_n linearly. For systems without their own table, the reference quantities are taken at the *equal-multiplicity centrality*[E]

$$c_{\text{eq}} = c_{\text{ref}}\Big(\ln\frac{\mathrm{d}N_{\text{ch}}}{\mathrm{d}\eta}\Big) \quad \text{(interpolation, clamped to the table)}, \tag{47}$$

and the flow targets follow linear response[E]:

$$v_n^{\text{target}} = v_n^{\text{ref}}(c_{\text{eq}})\,\frac{\varepsilon_n}{\varepsilon_n^{\text{ref}}(c_{\text{eq}})} \quad (n = 2, 3), \qquad v_4^{\text{target}} = v_4^{\text{ref}}(c_{\text{eq}}). \tag{48}$$

Stage 1 – flow strength. The gain g is adjusted until the midrapidity ($|y| < 0.5$) pion $\langle p_T\rangle$ matches data within 3%: $g \to \text{clip}\left(g\,(\langle p_T\rangle_{\text{data}}/\langle p_T\rangle_{\text{model}})^{1/2}, 0.3, 3\right)$ until the target is bracketed, then bisection $g \to \frac{1}{2}(g_{\text{lo}} + g_{\text{hi}})$ (at most 12 iterations).

Stage 2 – anisotropy. Start $a_{2,3} = 0.8\,v_{2,3}^{\text{target}}$, $a_4 = 0.5\,v_4^{\text{target}}$; iterate (at most 8 times)

$$a_n \to \text{clip}\left(a_n + 0.7\,(v_n^{\text{target}} - v_n^{\text{model}}),\ -0.8,\ 0.8\right) \quad \text{until} \quad |v_n^{\text{model}} - v_n^{\text{target}}| \le \max(0.003,\ 0.05|v_n^{\text{target}}|), \tag{49}$$

with v_n^{model} measured on $\ge 8\times10^4$ bootstrapped emissions of the event's own tracers. A TRENTo event file, when supplied, replaces T_R (bilinear resampling) and provides ε_n; Ψ_n are then computed from its grid with the field formula above.

9 Nucleons, spectators and the time axis

[hic_core/event.py]

Frame times[V]: N_{pre} approach frames and $N_{\text{post}} + 1$ collision frames with an optional time warp w_p,

$$t_i = -t_{\text{app}}\Big(1 - \frac{i}{N_{\text{pre}}}\Big), \qquad t_j = t_{\text{end}}\Big(\frac{j}{N_{\text{post}}}\Big)^{w_p}. \tag{50}$$

Nucleon positions: rest-frame z compressed by $\zeta = \max(1/\gamma,\ L_{\text{floor}}/2R)$ (the visual thickness floor L_{floor} is reported on screen[V]); before the collision ($t < 0$), with $d = \pm1$ the beam direction,

$$\vec{x}_\perp(t) = \vec{x}_{\perp,0}, \qquad z(t) = \zeta z_{\text{rest}} + d\,\beta\,t. \tag{51}$$

Spectators after the collision (beam rapidity plus Fermi motion, $p_F = 0.26$ GeV, Moniz et al. PRL 26 (1971) 445):

$$p_z = d\,\gamma\beta m, \quad p_{x,y} \sim \mathcal{N}\big(0,\, p_F^2/5\big), \quad E = \sqrt{p^2 + m^2}, \quad \vec{x}(t) = (\vec{x}_{\perp,0},\, \zeta z_{\mathrm{rest}}) + \frac{\vec{p}}{E}\, t. \tag{52}$$

Participants (visual hand-over to the fireball[V]): keep $z = \zeta z_{\mathrm{rest}} + d\beta t$ until $t = \tau_0 + 0.3$ fm/c, sphere radius scaled by $\max(0,\, 1 - t/(\tau_0 + 0.3))$.

10 Importers (Level B)

[hic_core/importers/; formats in docs/physics/04-simulation-formats.md]

Particles between output times move on straight lines (block at the latest $t_k \le t$); UrQMD final lists from each particle's own freeze-out point:

$$\vec{x}(t) = \vec{x}_k + \frac{\vec{p}}{E}\,(t - t_k), \qquad \vec{x}(t) = \vec{x}_f + \frac{\vec{p}}{E}\,(t - t_f),\ t \ge t_f. \tag{53}$$

Hydro grids are sampled in Milne coordinates, linearly in τ and trilinearly in (x, y, η_s), with fractional indices $f_x = (x - x_0)/\Delta x$ etc.:

$$T = (1 - w_\tau)\, T_k + w_\tau\, T_{k+1}, \qquad T_k = \sum_{a,b,c \in \{0,1\}} \lambda_x^a \lambda_y^b \lambda_\eta^c\, T_k[i_0 + a,\, j_0 + b,\, l_0 + c], \tag{54}$$

$\lambda^0 = 1 - \{f\}$, $\lambda^1 = \{f\}$; 2+1D grids apply for $|\eta_s| < \eta_{\mathrm{window}}$ (default 2). TRENTo cell centres and step: $x_i = (i + \frac{1}{2})\Delta - \frac{n\Delta}{2}$, $\Delta = \sqrt{\mathrm{mult}/\sum g}$ if not given (file stored $[i_y][i_x]$). vHLLE: $\tau_k = \tau_{k-1} + (\mathrm{step}_k - \mathrm{step}_{k-1})\,\Delta\tau$ (2 after the grid resize), origin after resize $= -(n-1)\Delta/2$. MUSIC: $\tau_k = \tau_0 + k\,\Delta\tau_{\mathrm{out}}$, $x_i = x_{\min} + i\,\Delta x$. Codes that output GeV: $T[\mathrm{MeV}] = 1000\,T[\mathrm{GeV}]$. Exporter (MUSIC format): $e = \varepsilon(T)$, $P = p(T)$, $c_s^2(T)$ from the EoS, cells with $e < e_{\mathrm{cut}}$ omitted.

11 Special functions

[hic_core/special.py; Abramowitz & Stegun 9.8.1–9.8.8; Numerical Recipes 6.2]

Regularised incomplete gamma: series for $x < a + 1$, Lentz continued fraction otherwise,

$$P(a, x) = \frac{e^{-x} x^a}{\Gamma(a)} \sum_{n \ge 0} \frac{x^n}{a(a+1)\cdots(a+n)}, \qquad Q = 1 - P, \tag{55}$$

$$Q(a, x) = \frac{e^{-x} x^a}{\Gamma(a)}\, \cfrac{1}{x + 1 - a - \cfrac{1 \cdot (1 - a)}{x + 3 - a - \cfrac{2 \cdot (2 - a)}{x + 5 - a - \cdots}}}. \tag{56}$$

Modified Bessel functions, $t = (x/3.75)^2$, $y = 3.75/x$, $u = (x/2)^2$, $w = 2/x$ (first line of each pair:

small argument, second: large argument):

$$I_0 = 1 + 3.5156229t + 3.0899424t^2 + 1.2067492t^3 + 0.2659732t^4 + 0.0360768t^5 + 0.0045813t^6 \qquad |x|<3.75$$
$$I_0 = \tfrac{e^x}{\sqrt{x}}\big(0.39894228 + 0.01328592y + 0.00225319y^2 - 0.00157565y^3 + 0.00916281y^4 - 0.02057706y^5 + 0.02635537y^6 - 0.01647633y^7 + 0.00392377y^8\big) \qquad x\geq 3.75 \tag{57}$$

$$I_1 = x\big(0.5 + 0.87890594t + 0.51498869t^2 + 0.15084934t^3 + 0.02658733t^4 + 0.00301532t^5 + 0.00032411t^6\big)$$
$$I_1 = \tfrac{e^x}{\sqrt{x}}\big(0.39894228 - 0.03988024y - 0.00362018y^2 + 0.00163801y^3 - 0.01031555y^4 + 0.02282967y^5 - 0.02895312y^6 + 0.01787654y^7 - 0.00420059y^8\big) \tag{58}$$

$$K_0 = -\ln\tfrac{x}{2}\, I_0 - 0.57721566 + 0.42278420u + 0.23069756u^2 + 0.03488590u^3 + 0.00262698u^4 + 0.00010750u^5 + 0.0000074u^6 \qquad x\leq 2$$
$$K_0 = \tfrac{e^{-x}}{\sqrt{x}}\big(1.25331414 - 0.07832358w + 0.02189568w^2 - 0.01062446w^3 + 0.00587872w^4 - 0.00251540w^5 + 0.00053208w^6\big) \qquad x>2 \tag{59}$$

$$K_1 = \ln\tfrac{x}{2}\, I_1 + \tfrac{1}{x}\big(1 + 0.15443144u - 0.67278579u^2 - 0.18156897u^3 - 0.01919402u^4 - 0.00110404u^5 - 0.00004686u^6\big) \qquad x\leq 2$$
$$K_1 = \tfrac{e^{-x}}{\sqrt{x}}\big(1.25331414 + 0.23498619w - 0.03655620w^2 + 0.01504268w^3 - 0.00780353w^4 + 0.00325614w^5 - 0.00068245w^6\big) \qquad x>2 \tag{60}$$

12 Visual mappings in Blender[V]

[hic_viz/scene.py]

These only decide how numbers are drawn.

Colour. Blackbody colour of a mapped temperature (the real temperatures are $\sim 2\times10^{12}$ K):

$$T_{\text{colour}}[\text{K}] = 13\,\tfrac{\text{K}}{\text{MeV}} \times T[\text{MeV}] \qquad (T_c \mapsto 2000\text{ K},\ 500\text{ MeV} \mapsto 6500\text{ K}). \tag{61}$$

Volume density of the fireball shader (smoothstep fade; $(T_1, T_2) = (156.5, 172)$ MeV when hadrons are drawn, $(110, 150)$ MeV otherwise):

$$\rho_{\text{vol}} = 0.15\sqrt{\frac{\varepsilon}{10\text{ GeV fm}^{-3}}}\ \text{smoothstep}(T; T_1, T_2),$$
$$\text{smoothstep} = u^2(3-2u), \qquad u = \text{clip}\left(\frac{T-T_1}{T_2-T_1}, 0, 1\right). \tag{62}$$

Sizes. Nucleon spheres 0.8 fm, flattened along the beam by $\max(1/\gamma,\, L_{\text{floor}}/1.6)$; hadron spheres 0.15 fm (fluid phase) and 0.25 fm (free streaming) times a species factor (0.8–1.25).

Framing. Regions that must stay in view (longitudinal L, transverse $T_\perp$; R = nuclear radius):

$$t<0:\quad L = |t| + 0.3R + 2, \qquad T_\perp = R + \tfrac{b}{2} + 1.5,$$
$$t\geq 0:\quad L = \max(R,\, 0.85t + 6), \qquad T_\perp = \max\big(R + \tfrac{b}{2} + 1.5,\ R + 1.5 + 0.6t\big). \tag{63}$$

Screen half-extents at distance d for the 40 mm lens / 36 mm sensor and aspect $a = W/H$:

$$(h_x, h_y) = \begin{cases} (0.45d,\ 0.45d/a) & a \geq 1 \\ (0.45d\,a,\ 0.45d) & a < 1 \end{cases},$$
$$\text{camera scale} = \max\left(\frac{L}{0.85\,h_\parallel},\ \frac{T_\perp}{0.85\,h_\perp}\right) \quad (\text{at } d = 50), \tag{64}$$

$h_\parallel, h_\perp$ being the screen half-extents along which the beam and the transverse plane are drawn. Orbit move: rotation angle $70^\circ \times u^2(3-2u)$ over the post-collision frames.

HUD. $t[\text{s}] = t[\text{fm}/c] \times 3.336\times10^{-24}$.

13 Validation metric

[scripts/validate.py]

$$\Delta = 100\,\frac{X_{\text{model}} - X_{\text{measured}}}{X_{\text{measured}}}\ \%. \tag{65}$$

14 Interactive explorer additions (The Little Bang)

[web/src/physics/*.js, web/src/scene/hits.js] The browser explorer ports Sections 1–8 (checked against Python in web/test/physics.test.mjs). It adds the following.

Custom nucleus (any A; Miller et al. 2007, nucl-ex/0701025; valley of stability)[E]:

$$R = 1.12\,A^{1/3} - 0.86\,A^{-1/3}\ \text{fm}, \qquad a = 0.54\ \text{fm}, \qquad Z = \text{round}\,\frac{A}{1.98 + 0.0155\,A^{2/3}}. \tag{66}$$

Optical-Glauber centrality (for custom nuclei; same review, Eq. 5):

$$T_A(s) = \int \rho_A\big(\sqrt{s^2+z^2}\big)\,dz, \quad \int T_A\,d^2s = A, \qquad T_{AB}(b) = \int T_A(\vec{s})\,T_B(\vec{s}-\vec{b})\,d^2s, \tag{67}$$

$$P_{\text{inel}}(b) = 1 - \Big(1 - \frac{\sigma_{NN}T_{AB}(b)}{AB}\Big)^{AB}, \qquad c(b) = \frac{\int_0^b P_{\text{inel}}\,2\pi b'\,db'}{\int_0^\infty P_{\text{inel}}\,2\pi b'\,db'}. \tag{68}$$

Coulomb deflection of the projectile (impact approximation; Jackson §13.1):

$$\theta_C = \frac{2Z_AZ_B\,\alpha\hbar c}{b\,p\,\beta}, \qquad p = A_A\,\gamma\beta\,m_N. \tag{69}$$

Detector hits: $N = 1.6\,dN_{\text{ch}}/d\eta$ particles in $|\eta| < 0.8$, azimuths from $dN/d\varphi \propto 1 + 2\sum_{n=2}^{4} v_n \cos n(\varphi - \Psi_n)$, hit position $z = r\sinh\eta$ on a barrel of radius r; measured $v_2 = \langle\cos 2(\varphi - \Psi_2)\rangle \pm \sqrt{1/(2N)}$. Arrival times[V].

Multi-nucleon transfer explorer (The Nucleon Exchange)

Implementation: web/mnt/src/physics/*.js; plan, research and validation: docs/mnt/. Units: fm, fm/c, MeV, $c = 1$; $e^2 = 1.439964$ MeV fm, $\hbar c = 197.327$ MeV fm, $m_u = 931.494$ MeV, 1 zs $= 10^{-21}$ s $= 299.79$ fm/c.

15 Nuclear input data

[web/mnt/src/physics/nuclides.js; AME2020, FRDM2012, WS4] Mass excess $\Delta(Z,N)$: AME2020 where measured or extrapolated, FRDM2012 elsewhere (IAEA RIPL-4 merge).

$$B(Z,N) = Z\,\Delta_{^1\text{H}} + N\,\Delta_n - \Delta(Z,N), \qquad S_n(Z,N) = B(Z,N) - B(Z,N-1), \tag{70}$$

$$Q_{gg} = \Delta_P + \Delta_T - \Delta_1 - \Delta_2. \tag{71}$$

16 Nucleus–nucleus potential

[potential.js; Błocki et al. 1977; Bass 1980; Dutt & Puri 2010] **Proximity 1977** (default):

$$V_N = 4\pi\gamma\, b\, \bar{C}\, \Phi(\xi), \quad \xi = \frac{r - C_1 - C_2}{b}, \quad b = 1 \text{ fm}, \quad \bar{C} = \frac{C_1 C_2}{C_1 + C_2}, \quad C_i = R_i\Big[1 - \Big(\frac{b}{R_i}\Big)^2\Big], \tag{72}$$

$$R_i = 1.28 A_i^{1/3} - 0.76 + 0.8 A_i^{-1/3}, \qquad \gamma = 0.9517\big[1 - 1.7826\, I^2\big] \text{ MeV fm}^{-2}, \quad I = \frac{N - Z}{A}, \tag{73}$$

$$\Phi(\xi) = \begin{cases} -\frac{1}{2}(\xi - 2.54)^2 - 0.0852(\xi - 2.54)^3 & -1 \le \xi \le 1.2511 \\ -3.437\, e^{-\xi/0.75} & \xi \ge 1.2511 \end{cases} \tag{74}$$

Below $\xi = -1$ the cubic turns over (an artefact); it is continued linearly with a wall $6(\xi{+}1.5)^2$ for $\xi < -1.5$[E]. **Bass 1980** (barrier heights, V_B in the page): $V_N = -\frac{R_1 R_2}{R_1 + R_2}\big[0.033\, e^{s/3.5} + 0.007\, e^{s/0.65}\big]^{-1}$, $R_i = R_s(1 - 0.98/R_s^2)$. **Coulomb**: $V_C = Z_1 Z_2 e^2/r$ for $r \ge R_C$, $\frac{Z_1 Z_2 e^2}{2R_C}(3 - r^2/R_C^2)$ inside, $R_C = 1.2(A_1^{1/3} + A_2^{1/3})$.

Neck thinning and elongation[E]: $V_N(r, d) = \big(1 - d/d_{\rm sc}\big)\, V_N(r - d)$ with the scission elongation from Viola systematics, $d_{\rm sc} = r_{\rm sc} - C_1 - C_2$, $r_{\rm sc} = e^2(A_1^{1/3} + A_2^{1/3})/0.755$ (i.e. TKE $= 0.755\, Z_1 Z_2/(A_1^{1/3} + A_2^{1/3}) + 7.3$ MeV).

Deformed nuclei ($A \ge 100$, $|\beta_2| \ge 0.15$, FRDM2012): $C_i \to C_i + R_i\big[\beta_2 Y_{20}(\cos\theta_i) + \beta_4 Y_{40}(\cos\theta_i)\big]$, averaged over random orientations with 3-point Gauss–Legendre in $\cos\theta_i \in [0, 1]$ (nodes 0.1127, 0.5, 0.8873; weights $5/18, 8/18, 5/18$)[E].

17 Classical trajectory with surface friction

[trajectory.js; Gross & Kalinowski 1978; Abe & Bouriquet 2003]

$$\dot r = \frac{p}{\mu}, \quad \dot p = -\partial_r V(r, d) + \frac{L^2}{\mu r^3} - K_r\, \dot r, \quad \dot\theta = \frac{L}{\mu r^2}, \quad \dot L = -\frac{K_\phi}{\mu}\,(L - L_{\rm st}), \tag{75}$$

$$K_r = K_r^0 \Big(\frac{\partial V_N}{\partial r}\Big)^2, \quad K_\phi = K_\phi^0 \Big(\frac{\partial V_N}{\partial r}\Big)^2 (1 + x), \tag{76}$$

$$x = \frac{5}{2}\Big(\frac{r - d}{C_1 + C_2}\Big)^2, \quad L_{\rm st} = L_0\, \frac{x}{1 + x}, \tag{77}$$

$$\dot d = F(s)\, \frac{d_{\rm eq} - d}{\tau_d}, \quad d_{\rm eq} = f_V\, g(x_m)\, d_{\rm sc}, \quad \tau_d = 4 \text{ zs (Li et al. 2006)}, \tag{78}$$

$$F(s) = \frac{1}{1 + e^{(s - s_C)/a_C}}, \quad s = r - d - C_1 - C_2. \tag{79}$$

$\mu = m_u A_1 A_2/(A_1 + A_2)$, $L_0 = \ell\hbar$, K^0 converted with 1 s $= 2.998 \times 10^{23}$ fm/c. Mean fissility (du Rietz et al. 2013) and quasi-fission propensity[E]:

$$x_m = \tfrac{3}{4} x_{\rm eff} + \tfrac{1}{4} x_{\rm CN}, \qquad g = \frac{1}{1 + e^{-(x_m - 0.79)/0.02}}, \tag{80}$$

$$x_{\rm eff} = \frac{4 Z_1 Z_2/\big(A_1^{1/3} A_2^{1/3}(A_1^{1/3} + A_2^{1/3})\big)}{50.883(1 - 1.7826 I^2)}, \quad x_{\rm CN} = \frac{Z^2/A}{50.883(1 - 1.7826 I^2)}. \tag{81}$$

Energies: $E_{\rm rel} = \frac{p^2}{2\mu} + \frac{L^2}{2\mu r^2} + V(r, d)$, $E_{\rm spin} = \frac{(L_0 - L)^2}{2(\mathcal{J}_1 + \mathcal{J}_2)}$, $\mathcal{J}_i = \frac{2}{5} m_i C_i^2$, $E_{\rm diss} = E_{\rm cm} - E_{\rm rel} - E_{\rm spin}$, TKEL $= E_{\rm diss} + E_{\rm spin}$. **Deflection function** with the Coulomb tails ($k = Z_1 Z_2 e^2$):

$$\Theta(\ell) = \pi - \big(\Delta\theta_{\rm in} + \Delta\theta_{\rm orbit} + \Delta\theta_{\rm out}\big), \tag{82}$$

$$\Delta\theta(E, L, r) = \arccos\frac{1}{\varepsilon} - \arccos\frac{L^2/(\mu k r) + 1}{\varepsilon}, \quad \varepsilon = \sqrt{1 + \frac{2EL^2}{\mu k^2}}, \tag{83}$$

Rutherford when the orbit never enters $r_0 = 60$ fm: $\sin(\Theta/2) = 1/\varepsilon$.

Capture and quasi-fission. Trapped when $E_{\rm rel} < V_{\rm top}$, the outer barrier of $V_{\rm eff} = V(r', d) + L^2/(2\mu r'^2)$. At first capture a fraction $1 - g(x_m)$ fuses[E]; the rest decays with the Kramers rate (Zhu, Feng & Zhang 2015: $\Gamma = 2.8$, $\hbar\omega_B = 2.0$, $\hbar\omega = 3.0$ MeV)

$$\Lambda_{\rm qf} = \frac{\omega}{2\pi\omega_B}\left[\sqrt{(\Gamma/2\hbar)^2 + \omega_B^2} - \Gamma/2\hbar\right] e^{-B_{\rm qf}/T}, \tag{84}$$

$$B_{\rm qf} = V_{\rm top} - E_{\rm rel}, \quad T = \sqrt{E_{\rm diss}/a}, \; a = A/12 \text{ MeV}^{-1}, \tag{85}$$

leaving with TKE $= V_{\rm top}$ at the angle reached at that time; systems still together after 15 zs count as captured[E].

18 Nucleon exchange

[`transfer.js`; Feng et al. 2009; Ayik et al. 2018] Master equation for the projectile-like fragment (Z_1, N_1) with single-nucleon steps and bounded detailed-balance rates[E]:

$$\frac{dP_i}{dt} = \sum_{j \in \mathrm{nb}(i)} \left(W_{j\to i} P_j - W_{i\to j} P_i\right), \tag{86}$$

$$W_{i\to j} = \frac{2D_\alpha}{1 + e^{(U_j - U_i)/T_{\rm eff}}}, \quad \frac{W_{i\to j}}{W_{j\to i}} = e^{-(U_j - U_i)/T_{\rm eff}}, \tag{87}$$

$$D_{NN} = D^0_{NN}\, F(s(t)), \quad D_{ZZ} = \tfrac{1}{2} D_{NN}, \quad T_{\rm eff} = \max(T, T_{\min}). \tag{88}$$

For $|\Delta U| \ll T$ the drift is $-D\,\partial U/T$ (Einstein relation). Driving potential

$$U(Z_1, N_1) = \Delta_1 + \Delta_2 - \Delta_P - \Delta_T - (1 - \phi_1) E_{\rm mic,1} - (1 - \phi_2) E_{\rm mic,2} \tag{89}$$

$$+ \frac{Z_1 Z_2 e^2}{R} + \frac{\hbar^2 \ell(\ell+1)}{2\mathcal{J}_{\rm rel}} - U_{\rm entrance}, \tag{90}$$

$$\phi(T, A) = \frac{\tau}{\sinh \tau}, \quad \tau = \frac{2\pi^2 T}{41 A^{-1/3}} \quad \text{(Bohr \& Mottelson; Ignatyuk 1975)}, \tag{91}$$

$$R = C(A_1) + C(A_2) + 1 \text{ fm}, \quad \mathcal{J}_{\rm rel} = \mu R^2 + \tfrac{2}{5} m_u \left(A_1 C_1^2 + A_2 C_2^2\right). \tag{92}$$

$E_{\rm mic}$ is the FRDM2012 microscopic (shell + pairing) energy.

19 Cross sections, kinematics and de-excitation

[`model.js`, `evaporation.js`, `compare.js`]

$$\sigma_{\rm pr}(Z_1, N_1) = \frac{\pi}{k^2} \sum_\ell (2\ell + 1) \sum_{\rm exits} w\, P_\ell(Z_1, N_1), \tag{93}$$

$$k^2 = \frac{2\mu E_{\rm cm}}{\hbar^2}, \tag{94}$$

$$E^* = E_{\rm diss} + Q_{gg}, \qquad E^*_i = E^* \frac{A_i}{A}, \qquad \text{TKE} = E_{\rm cm} + Q_{gg} - E^*. \tag{95}$$

Neutron evaporation: while $E^* > S_n$, emit with spectrum $\varepsilon\, e^{-\varepsilon/T}$, $T = \sqrt{E^*/a}$, $a = A/12$; for $Z \geq 88$ compete with fission (Yanez & Loveland 2015):

$$\frac{\Gamma_n}{\Gamma_f} = \frac{4A^{2/3} a_f (E^* - S_n)}{K_0\, a_n \left[2\sqrt{a_f (E^* - B_f)} - 1\right]} \tag{96}$$

$$\times \exp\!\left[2\sqrt{a_n (E^* - S_n)} - 2\sqrt{a_f (E^* - B_f)}\right], \quad K_0 = 14.4 \text{ MeV}, \; a_n = a_f, \tag{97}$$

$$B_f(E^*) = B_{\mathrm{LD}} + (B_f^{\mathrm{WS4}} - B_{\mathrm{LD}})\, e^{-E^*/18.5\,\mathrm{MeV}}\text{[E]}, \tag{98}$$

$$B_{\mathrm{LD}} = \begin{cases} 0.38(0.75 - x) E_s & x < \tfrac{2}{3} \\ 0.83(1 - x)^3 E_s & x \geq \tfrac{2}{3} \end{cases}, \quad E_s = 17.9439(1 - 1.7826 I^2) A^{2/3}. \tag{99}$$

Lab angle of the projectile-like fragment (detector windows): $\tan\theta_{\mathrm{lab}} = \sin\chi / (\cos\chi + V_{\mathrm{cm}}/v_1)$, $\chi = \arccos(\cos\Theta)$, $v_1 = \sqrt{2\,\mathrm{TKE}\, A_T / (A_P (A_P + A_T))}$, accepted fraction of a Gaussian of width 4° inside the window[E].

Calibration and validation (`scripts/mnt_validate.mjs`): global parameters $(K_r^0, D_{NN}^0, s_C, a_C, T_{\min}) = (2 \times 10^{-23}$ s/MeV, 0.06 $(\mathrm{fm}/c)^{-1}$, 1.5 fm, 0.4 fm, 7 MeV)[E], fitted once by grid search on four training datasets (minimum mean rms of tef) and frozen;

$$\mathrm{tef} = \log_{10} \frac{\sigma_{\mathrm{model}}}{\sigma_{\mathrm{exp}}} \quad \text{per measured nuclide (Barrett et al. 2015).} \tag{100}$$

Thick targets: model averaged at 3 energies across the target. $K_\phi^0 = 0.01 \times 10^{-23}$ s/MeV (Gross–Kalinowski).

Visual mappings[V]: nucleons placed uniformly in spheres of radius $1.12A^{1/3}$ (hard core 1.5 fm); nucleons move across the neck when round$\langle Z_1\rangle$ or round$\langle N_1\rangle$ changes; extra swaps at rate $0.25\,F$ per frame; neck radius $0.8 + 2.6F$ fm; Wilczyński smearing $\sigma_\theta = 3^\circ + 0.04\chi$, $\sigma_E = 3 + 0.08\,\mathrm{TKEL}$ MeV.